

0620 • PAPER 4

TOPIC

10

Chemistry of the environment

Questions only • Original examination pages retained

14

QUESTIONS

180

TOTAL MARKS

1

SECTIONS



USE THE TRACKER • ATTEMPT WITHOUT NOTES • MARK • REVIEW ERRORS

TOPIC CONTENTS

Quick navigation and progress overview

1/1

01

Air, water, climate and industrial chemistry

14 questions • 180 marks

10.1 Air, water, climate and industrial chemistry

#	SOURCE	MARKS	DONE	MARK	REVIEW
01	2020 May/June Paper 41 Question 3	17	☐	☐	☐
02	2020 May/June Paper 42 Question 3	16	☐	☐	☐
03	2020 May/June Paper 43 Question 4	8	☐	☐	☐
04	2020 Oct/Nov Paper 41 Question 4	10	☐	☐	☐
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06	2021 Oct/Nov Paper 41 Question 4	12	☐	☐	☐
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14	2024 May/June Paper 42 Question 1	7	☐	☐	☐

3 Sulfur dioxide, SO_2 , is used in the manufacture of sulfuric acid.

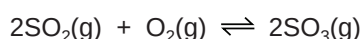
(a) In the first stage of the process, sulfur dioxide is obtained from sulfur-containing ores.

Name **one** of these ores.

..... [1]

(b) The next stage of the process is a reaction which can reach equilibrium.

The equation for this stage is shown.



(i) Describe **two** features of an equilibrium.

.....
..... [2]

(ii) Name the catalyst used in this stage.

..... [1]

(iii) Why is a catalyst used?

..... [1]

(iv) Explain, in terms of particles, why a high temperature increases the rate of this reaction.

.....
.....
.....
.....
..... [3]

(v) In this stage, only a moderate temperature of 450°C is used.

What does this suggest about the forward reaction?

..... [1]

(vi) Calculate the percentage by mass of sulfur in sulfur trioxide, SO_3 .

percentage = [2]

- (c) Concentrated sulfuric acid is a dehydrating agent which can chemically remove water from substances.

Both hydrated copper(II) sulfate crystals and sucrose (a sugar), $C_{12}H_{22}O_{11}$, can be completely dehydrated by concentrated sulfuric acid.

Name the solid product formed in each case.

hydrated copper(II) sulfate crystals

sucrose [2]

- (d) When propan-1-ol is heated with concentrated sulfuric acid as a catalyst an unsaturated hydrocarbon of relative molecular mass 42 is formed and one other product.

- (i) What is meant by the term *unsaturated*?

..... [1]

- (ii) Write the chemical equation for this reaction.

..... [2]

- (iii) Name the unsaturated hydrocarbon formed.

..... [1]

[Total: 17]

3 (a) Sulfuric acid is made from sulfur in a four-stage process.

stage 1 Sulfur is converted into sulfur dioxide.

stage 2 Sulfur dioxide is converted into sulfur trioxide.

stage 3 Sulfur trioxide is converted into oleum.

stage 4 Oleum is converted into sulfuric acid.

(i) How is sulfur converted into sulfur dioxide in **stage 1**?

..... [1]

(ii) Describe how sulfur dioxide is converted into sulfur trioxide in **stage 2**.

Your answer should include:

- an equation for the reaction
- the temperature used
- the name of the catalyst used.

.....
.....
.....
..... [3]

(iii) The reaction in **stage 2** can reach equilibrium.

What is meant by the term *equilibrium*?

.....
.....
..... [2]

(b) Sulfur trioxide is converted into oleum, $\text{H}_2\text{S}_2\text{O}_7$, in **stage 3**.

What is sulfur trioxide reacted with to convert it into oleum?

..... [1]

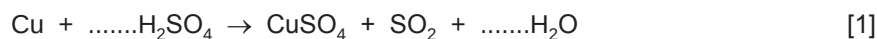
(c) Oleum is converted into sulfuric acid in **stage 4**.

Write a chemical equation for the conversion of oleum, $\text{H}_2\text{S}_2\text{O}_7$, into sulfuric acid.

..... [2]

- (d) When copper is reacted with hot concentrated sulfuric acid, sulfur dioxide gas is formed.

Balance the chemical equation for this reaction.



- (e) Sulfur dioxide is a reducing agent.

Give the colour change that occurs when excess sulfur dioxide is bubbled into acidified aqueous potassium manganate(VII).

starting colour of the solution

final colour of the solution [1]

- (f) When sulfuric acid reacts with ammonia the salt produced is ammonium sulfate.

Write the chemical equation for this reaction.

..... [2]

- (g) Barium sulfate is an insoluble salt.

Barium sulfate can be made from aqueous ammonium sulfate using a precipitation reaction.

- (i) Name a solution that can be added to aqueous ammonium sulfate to produce a precipitate of barium sulfate.

..... [1]

- (ii) Write an ionic equation for this precipitation reaction. Include state symbols.

..... [2]

[Total: 16]

- 4 (a) Filtration and chlorination are two stages in water treatment.

State the purpose of each stage.

filtration
.....
chlorination
..... [2]

- (b) A student uses anhydrous copper(II) sulfate to test for the presence of water.

- (i) What colour change is seen if water is present?

from to [2]

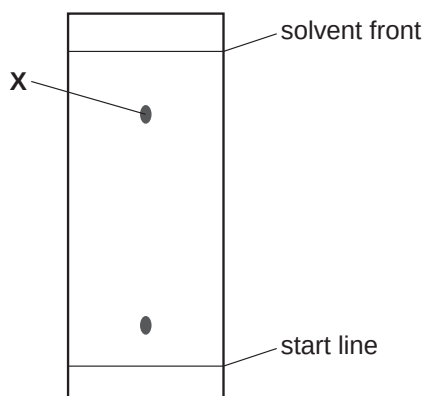
- (ii) The purity of a sample of water can be assessed by measuring its boiling point.

How is the boiling point of water affected by impurities?

..... [1]

- (c) Chromatography can be used to test the purity of substances.

The diagram shows the chromatogram of a coloured substance.



- (i) How does this chromatogram show that this substance is **not** pure?

..... [1]

- (ii) Draw a circle round the correct R_f value for the spot labelled X.

0.2 0.4 0.8 1.2 [1]

- (iii) State how a colourless substance can be made visible on a chromatogram.

..... [1]

[Total: 8]

4 Air is a mixture of gases.

(a) State the percentage of clean dry air which is oxygen. Give your answer to the nearest whole number.

..... % [1]

(b) Oxygen and nitrogen are useful gases that can be obtained from air.

(i) Name the process used to separate oxygen and nitrogen from liquid air.

..... [2]

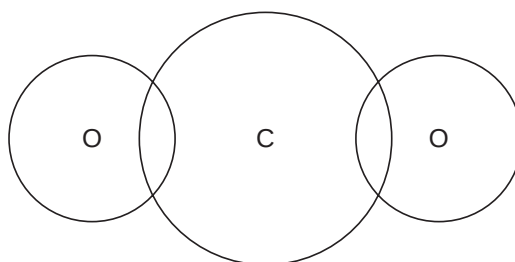
(ii) State the property of oxygen and nitrogen that allows these gases to be separated using this process.

..... [1]

(c) Carbon dioxide, CO_2 , is a covalent molecule.

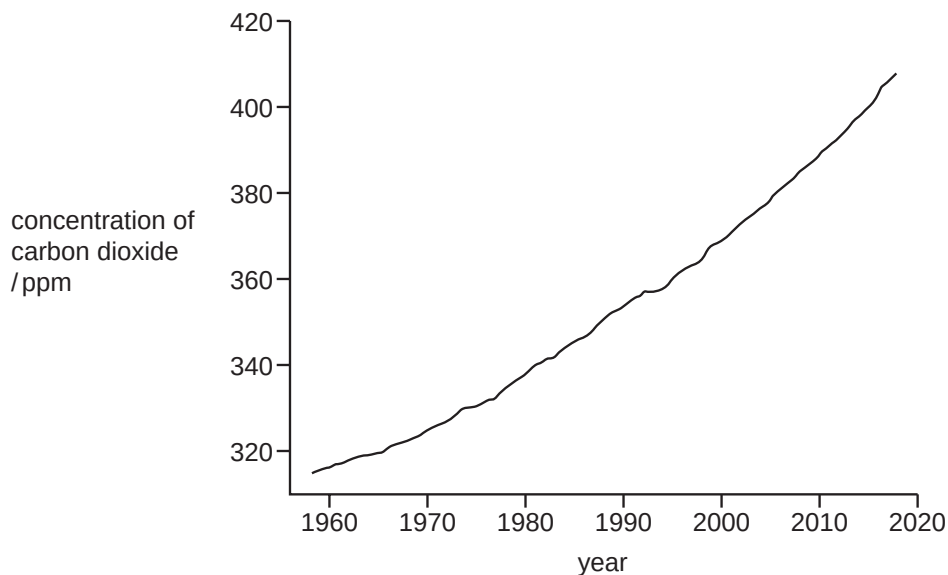
Complete the diagram to show the electron arrangement in one molecule of CO_2 .

Show only the outer electrons.



[2]

- (d) The graph shows the concentration of carbon dioxide in the atmosphere over a 60-year period, measured in parts per million (ppm).



The data shown in the graph is of global concern.

Explain why.

.....

.....

.....

.....

.....

..... [3]

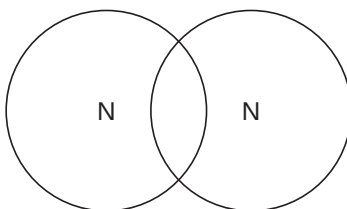
- (e) Name the process in the carbon cycle by which plants remove carbon dioxide from the atmosphere.

..... [1]

[Total: 10]

3 This question is about nitrogen and some of its compounds.

- (a) Complete the dot-and-cross diagram to show the electron arrangement in a molecule of nitrogen, N_2 .
Show the outer shell electrons only.



[2]

- (b) Nitrogen can be converted into ammonia by the Haber process.

- (i) Describe how nitrogen is obtained for the Haber process.

.....
..... [2]

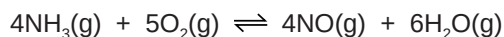
- (ii) Give the essential reaction conditions and write a chemical equation for the reaction occurring in the Haber process.

chemical equation:
.....
reaction conditions:
.....
.....
.....
.....

[5]

- (c) Some of the ammonia made by the Haber process is converted into nitric acid.

The first stage of this process is the oxidation of ammonia to make nitrogen monoxide.



The process is carried out at 900 °C and a pressure of 5 atmospheres using an alloy of platinum and rhodium as a catalyst.

The forward reaction is exothermic.

- (i) State the meaning of the term *catalyst*.

.....
 [2]

- (ii) State the meaning of the term *oxidation*.

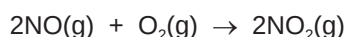
..... [1]

- (iii) Complete the table using the words **increase**, **decrease** or **no change**.

	effect on the rate of the forward reaction	effect on the equilibrium yield of NO(g)
increasing the temperature		
increasing the pressure		

[4]

- (d) Nitrogen monoxide, NO, is converted into nitrogen dioxide, NO₂.



The nitrogen dioxide reacts with oxygen and water to produce nitric acid as the only product.

Write a chemical equation for this reaction.

..... [2]

(e) Ammonium nitrate, NH_4NO_3 , is a fertiliser.

Calculate the percentage by mass of nitrogen in ammonium nitrate.

..... % [2]

[Total: 20]

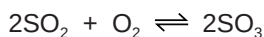
4 Chalcopyrite, FeCuS_2 , is used in the manufacture of sulfuric acid in the Contact process.

- (a) In the first stage of the process, chalcopyrite reacts with oxygen in the air to produce sulfur dioxide, SO_2 , iron(III) oxide and copper(II) oxide.

Complete the chemical equation for the reaction of FeCuS_2 with oxygen.



- (b) Sulfur dioxide is then converted to sulfur trioxide.



The reaction is exothermic. It is also an equilibrium.

- (i) State **two** features of an equilibrium.

1

2 [2]

- (ii) State the temperature and pressure used in this reaction.
Include units.

• temperature

• pressure [2]

- (iii) Name the catalyst used.

..... [1]

- (iv) Explain why a catalyst is used.

..... [1]

- (v) Describe and explain, in terms of equilibrium, what happens when the temperature is increased.

.....

..... [2]

- (c) Concentrated sulfuric acid is a dehydrating agent.

When glucose is dehydrated, carbon and one other product are formed.

Complete the equation to show the dehydration of glucose, $\text{C}_6\text{H}_{12}\text{O}_6$.



[Total: 12]

- 1 A list of substances is shown.

ammonia
bauxite
carbon dioxide
carbon monoxide
ethanol
hematite
oxygen
sodium chloride
sulfur dioxide

Answer the questions using the list of substances.

Each substance may be used once, more than once or not at all.

State which substance is:

- (a) an element [1]
- (b) an ore of aluminium [1]
- (c) a gas that causes acid rain [1]
- (d) used as a fuel [1]
- (e) an ionic compound [1]
- (f) produced in the Haber process [1]
- (g) a product of respiration [1]
- (h) a toxic product of the incomplete combustion of hydrocarbons
..... [1]
- (i) a gas produced in the test for nitrate ions. [1]

[Total: 9]

4 This question is about compounds of sulfur.

- (a) Sulfuric acid, H_2SO_4 , is manufactured using the Contact process. This manufacture involves four stages.

stage 1 Molten sulfur burns in air to produce sulfur dioxide.

stage 2 Sulfur dioxide reacts with oxygen to form sulfur trioxide.

stage 3 Sulfur trioxide combines with concentrated sulfuric acid to form oleum, $\text{H}_2\text{S}_2\text{O}_7$.

stage 4 Oleum reacts to form concentrated sulfuric acid.

- (i) Write a chemical equation for the reaction occurring in **stage 1**.

..... [1]

- (ii) State the essential conditions that are necessary for **stage 2**. Write an equation for the chemical reaction that occurs.

.....
.....
.....
..... [4]

- (iii) Write a chemical equation for the reaction occurring in **stage 3**.

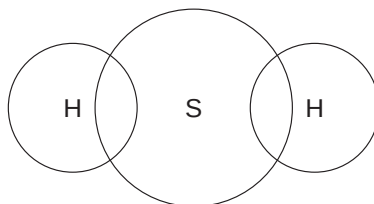
..... [1]

- (iv) Name the substance that reacts with oleum in **stage 4**.

..... [1]

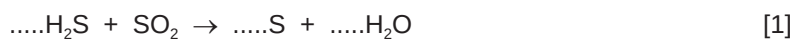
(b) Hydrogen sulfide has the formula H_2S .

- (i) Complete the dot-and-cross diagram to show the electron arrangement in a molecule of hydrogen sulfide. Show outer shell electrons only.



[2]

- (ii) Balance the chemical equation for the reaction of hydrogen sulfide with sulfur dioxide shown.



[Total: 10]

1 This question is about gases found in clean, dry air and gases found in polluted air.

(a) Name **one** gas found in clean, dry air which contributes to global warming.

..... [1]

(b) State the percentage of nitrogen in clean, dry air.

..... [1]

(c) Name the substance used to remove sulfur dioxide in flue gas desulfurisation.

..... [1]

(d) Nitrogen dioxide, NO_2 , is formed in car engines.

Name the equipment in a car exhaust used to remove the NO_2 formed in car engines.

..... [1]

(e) All gases diffuse.

(i) Choose from the list of formulae the gas which diffuses most quickly.

Draw a circle around your answer.

CO CO₂ CH₄ NO₂ SO₂

[1]

(ii) Explain your answer to (i).

..... [1]

(f) State **one** adverse effect of carbon monoxide on human health.

..... [1]

(g) Carbon dioxide, CO_2 , is a reactant in photosynthesis.

Name the **two** products of photosynthesis.

..... and [2]

(h) Complete the dot-and-cross diagram in Fig. 1.1 for a molecule of CO_2 .

Show outer shell electrons only.

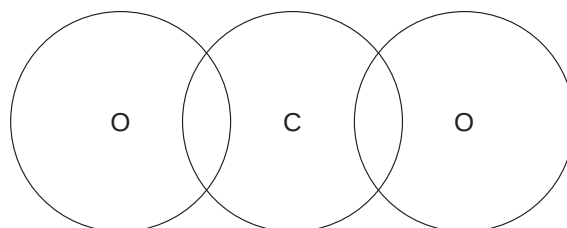


Fig. 1.1

[2]

[Total: 11]

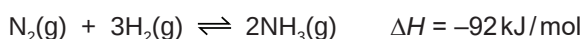
3 The Haber process is used to manufacture ammonia.

(a) State the main source of each gas used in the Haber process.

nitrogen

hydrogen [2]

(b) The equation for the Haber process is shown.



The reaction is reversible. The forward reaction is exothermic.

(i) State what is meant by the symbol ΔH .

..... [1]

(ii) ΔH for the forward reaction is -92 kJ/mol .

State why this value shows that the forward reaction is exothermic.

..... [1]

(iii) State the typical conditions and name the catalyst used in the Haber process.

temperature °C

pressure kPa

catalyst [3]

(iv) Complete Table 3.1 to show the effect, if any, when the typical conditions in the Haber process are changed. Use only the words **increases**, **decreases** or **no change**.

Table 3.1

change to typical conditions	effect on the rate of the forward reaction	effect on the concentration of $\text{NH}_3(\text{g})$ at equilibrium
temperature increases	increases	
pressure decreases		
no catalyst	decreases	

[4]

- (v) Explain in terms of collision theory why increasing the temperature increases the rate of the reaction.

.....
.....
.....
.....
..... [3]

- (c) Ammonia reacts with an acid to form ammonium sulfate, $(\text{NH}_4)_2\text{SO}_4$.

- (i) State the formula of the acid used.

..... [1]

- (ii) State **one** use of ammonium sulfate.

..... [1]

- (iii) Calculate the percentage composition by mass of nitrogen in $(\text{NH}_4)_2\text{SO}_4$.

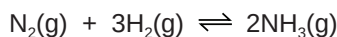
percentage of nitrogen = % [2]

[Total: 18]

6 This question is about nitrogen and compounds of nitrogen.

(a) Ammonia is manufactured by the reaction between nitrogen and hydrogen in the Haber process.

The equation is shown.



(i) State the source of nitrogen for the Haber process.

..... [1]

(ii) State the source of hydrogen for the Haber process.

..... [1]

(iii) State the typical conditions used in the Haber process.

temperature °C

pressure atm

[2]

(iv) Name the catalyst used in the Haber process.

..... [1]

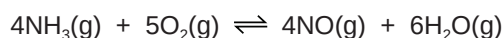
(v) State what is meant by the term catalyst.

.....

..... [2]

(b) Ammonia is converted into nitric acid.

(i) The first stage is the conversion of ammonia into nitrogen monoxide, NO.
The equation is shown.



The reaction is carried out at a temperature of 900 °C and a pressure of 7 atm.
The forward reaction is exothermic.

Using explanations that do **not** involve cost:

- explain why a temperature less than 900 °C is **not** used

.....

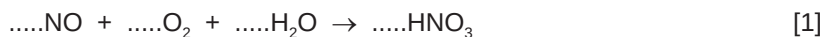
- explain why a pressure greater than 7 atm is **not** used.

.....

[2]

- (ii) In the second stage, nitrogen monoxide reacts with water and oxygen to produce nitric acid.

Balance the symbol equation for the reaction.



- (c) A student makes aqueous copper(II) nitrate by adding an excess of solid copper(II) carbonate to dilute nitric acid.

- (i) Write the symbol equation for this reaction.

..... [2]

- (ii) State **two** observations that indicate the copper(II) carbonate is in excess.

1

2 [2]

- (iii) Name **one** compound, other than copper(II) carbonate, that can be added to dilute nitric acid to produce aqueous copper(II) nitrate.

..... [1]

[Total: 15]

- 1 A list of gases is shown.

ammonia
carbon dioxide
carbon monoxide
ethene
fluorine
oxygen
sulfur dioxide
xenon

Answer the following questions using only the gases from the list.
Each gas may be used once, more than once or not at all.

Give the name of the gas that:

- (a) causes acid rain

..... [1]

- (b) forms an alkaline solution when dissolved in water

..... [1]

- (c) is inert

..... [1]

- (d) is a product of photosynthesis

..... [1]

- (e) can form a polymer

..... [1]

- (f) is produced in the test for nitrate ions.

..... [1]

[Total: 6]

3 This question is about nitrogen and some of its compounds.

(a) Nitrogen is converted into ammonia, NH_3 , in the Haber process.

(i) Nitrogen is obtained from air.

State the percentage of nitrogen in clean, dry air.

..... [1]

(ii) State the source of hydrogen for the Haber process.

..... [1]

(iii) Complete the dot-and-cross diagram in Fig. 3.1 for a molecule of ammonia.

Show the outer shell electrons only.

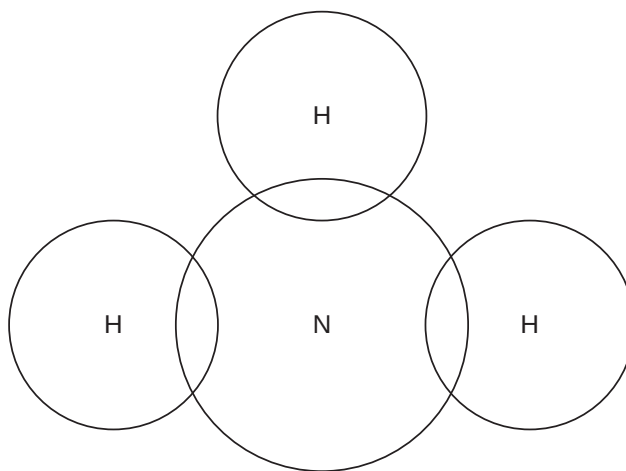


Fig. 3.1

[2]

(iv) Write a chemical equation for the reaction occurring in the Haber process and give the typical reaction conditions. Include units where appropriate.

chemical equation

reaction conditions:

temperature

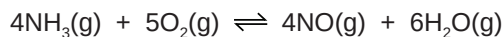
pressure

catalyst

[5]

(b) Ammonia is converted into nitric acid.

The first stage of this conversion uses a catalyst and occurs at a temperature of 900 °C and a pressure of 5 atmospheres.



The forward reaction is exothermic.

(i) Suggest which of the following elements is most likely to be used as a catalyst. Draw a circle around your answer.

calcium lead platinum sodium sulfur [1]

(ii) State the oxidation number of nitrogen in:

NH₃

NO [2]

(iii) Use your answer to (ii) to explain whether the nitrogen in ammonia undergoes oxidation or reduction.

..... [1]

(iv) Complete Table 3.1 using the words **increases**, **decreases** or **no change**.

Table 3.1

	effect on the equilibrium yield of NO(g)	effect on the rate of the forward reaction
decreasing the pressure		
decreasing the temperature		decreases
removing the catalyst		decreases

[4]

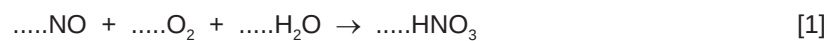
(v) Decreasing the temperature causes the rate of the forward reaction to decrease.

Explain, using collision theory, why the rate of the reaction is slower at the decreased temperature.

..... [3]

(c) In the second stage, nitric acid is produced.

Balance the symbol equation for this reaction.



[Total: 21]

1 A list of gases is shown.

ammonia
helium
hydrogen
carbon dioxide
carbon monoxide
chlorine
methane
nitrogen dioxide
propene
sulfur dioxide

Answer the following questions about these gases.

Each gas may be used once, more than once or not at all.

(a) State **one** gas which:

(i) is the main constituent of natural gas

..... [1]

(ii) is responsible for both photochemical smog **and** acid rain

..... [1]

(iii) is unsaturated

..... [1]

(iv) has monatomic particles

..... [1]

(v) reduces iron(III) oxide in a blast furnace.

..... [1]

(b) Nitrogen dioxide, NO_2 , and carbon monoxide are removed from a car exhaust by a catalytic converter.

Write the symbol equation for this reaction.

..... [2]

[Total: 7]

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